

Micro-tricks for Chapter 3 proofs

Statistics for Insurance the small moves that appear in almost every calculation

Each trick has the rule, why it holds, and where it was used in the material (Ex = exercises in the notes, MP = mock exam problems, EE = exam-style exercises PDF, PB = Potter guide).

A. The asymptotic language

What the symbols mean (as $x \rightarrow \infty$, or $h \rightarrow 0$)

$f \sim g$ $f/g \rightarrow 1$. “Same size, including the constant.” $2\bar{F}(z) - \bar{F}(z)^2 \sim 2\bar{F}(z)$.

$f = o(g)$ $f/g \rightarrow 0$. “Negligible compared with g .” $\bar{F}(z)^2 = o(\bar{F}(z))$. $o(1)$ = “something tending to 0”.

$f = O(g)$ $|f| \leq C|g|$ eventually. “At most the size of g .” $\sin x = O(1)$.

$f \approx g$ informal version of $\sim$. In a proof, replace it by $\sim$ or $\rightarrow$.

Arithmetic rules (as $h \rightarrow 0$): $O(h) + O(h^2) = O(h)$; $O(h) \cdot O(h) = O(h^2)$; $c \cdot O(h) = O(h)$; $o(1) \cdot O(1) = o(1)$; $A(1 + o(1)) \cdot B(1 + o(1)) = AB(1 + o(1))$. **Key rule:** if $f = g(1 + o(1))$ then $f \sim g$. That is why we write the main term times $(1 + o(1))$.

B. Expansions: replace a complicated function by its first-order term

Trick 1: First-order Taylor expansions for small h

$$(1+h)^a = 1 + ah + O(h^2), \quad \log(1+h) = h + O(h^2),$$

$$e^h = 1 + h + O(h^2), \quad \frac{1}{1+h} = 1 - h + O(h^2).$$

Why: the Taylor formula $f(h) = f(0) + f'(0)h + \frac{1}{2}f''(\xi)h^2$, with f'' bounded near 0.

Used in:

- **MP1 (Lomax bias):** $(1+h)^{-\alpha} = 1 - \alpha h + O(h^2)$ gave the bias $\frac{1}{\alpha+1}(k/n)^\gamma$.
- **Fréchet is RV** (Ex. 3.1.4): $1 - e^{-y} = y + o(y)$ with $y = x^{-1/\gamma}$, so $\ell(x) \rightarrow 1$.
- **Maxima** (next trick): $\log(1 - y/n) = -y/n + O(n^{-2})$.

Tip: first make the small quantity explicit, e.g. write $\frac{1+ut}{1+u} = t(1+h)$ with $h \rightarrow 0$. Then expand.

Trick 2: $(1 - y_n/n)^n \rightarrow e^{-y}$ whenever $y_n \rightarrow y$

Why: take logs: $n \log(1 - y_n/n) = n(-\frac{y_n}{n} + O(n^{-2})) = -y_n + O(1/n) \rightarrow -y$. Then exponentiate (exp is continuous).

Used in: MP3 (all maxima limits): $\mathbb{P}(M_n \leq a_n x) = F(a_n x)^n = (1 - \frac{n\bar{F}(a_n x)}{n})^n \rightarrow e^{-x^{-\alpha}}$ (Fréchet), and $(1 - \frac{e^{-x}}{n})^n \rightarrow e^{-e^{-x}}$ (Gumbel). **Pattern:** $F^n = (1 - \bar{F})^n$, so write $\bar{F}$ as “something/ n ”.

Trick 3: Binomial expansion for “at least one of n ”

$$1 - (1 - p)^n = np - \binom{n}{2}p^2 + \dots = np(1 + O(p)) \text{ as } p \rightarrow 0.$$

Used in: The single big event: $\mathbb{P}(M_2 > z) = 1 - (1 - \bar{F})^2 = 2\bar{F} - \bar{F}^2 \sim 2\bar{F}$; **MP2.1:** $\mathbb{P}(M_n > z) \sim n\bar{F}(z)$.

C. Comparing sizes: who wins?**Trick 4: The growth hierarchy**

As $x \rightarrow \infty$, each item is eventually negligible compared with the next:

$$\log \log x \ll \log x \ll (\log x)^2 \ll x^\varepsilon \ll x^p \ll e^{x^a} \ (a < 1) \ll e^x \ll e^{x^2}.$$

Consequences: $\log x \leq C_\varepsilon x^\varepsilon$ for any $\varepsilon > 0$; $x^p e^{-x^a} \rightarrow 0$; $e^{tx} x^{-\beta} \rightarrow \infty$; $\ell(x) x^{-\varepsilon} \rightarrow 0$ for slowly varying ℓ .

Why: take logs of the ratio. For example $\log(x^p/e^{x^a}) = p \log x - x^a \rightarrow -\infty$.

Used in: **RV** $\Rightarrow$ **heavy** ($e^{tx} \cdot x^{-\alpha-\varepsilon} \rightarrow \infty$); **Weibull**, **Ex 4** (tx vs x^a); **EE6:** $e^{tx-(\log x)^2} \rightarrow \infty$; **Potter** (powers beat ℓ); **PB Use 3** (“logs are free”).

Trick 5: Divide by the dominant term (factor it out)

In a sum, pull out the biggest piece. What remains tends to a constant:

$$1 + (x/\theta)^\eta = (x/\theta)^\eta (1 + (\theta/x)^\eta), \quad p x^{-2} + (1-p)x^{-3} = x^{-2} (p + (1-p)x^{-1}).$$

Used in: **Ratio test / RV proofs:** Burr (Ex 5), Lomax, log-logistic, mixtures; **max of two RV** (MP-style: divide by the heavier $\bar{F}_Z$); **lognormal** ratio in Ex. 3.2.

Trick 6: Ratio $\times$ vanishing factor

To show $A(x)^2/B(x) \rightarrow 0$, write it as $\frac{A}{B} \cdot A$: a bounded ratio times something $\rightarrow 0$.

Used in: **Ex 1A:** $\frac{\bar{F}(\delta x)^2}{\bar{F}(x)} = \underbrace{\frac{\bar{F}(\delta x)}{\bar{F}(x)}}_{\rightarrow \delta^{-\alpha}} \cdot \underbrace{\bar{F}(\delta x)}_{\rightarrow 0}$; **MP2.4(iii)**; **MP5.2** ($\frac{\bar{F}(u_n s) \bar{F}(u_n t)}{\bar{F}(u_n)} \rightarrow s^{-\alpha} \cdot 0$).

Trick 7: Telescoping a ratio into known limits

Insert factors that cancel so that each piece has a known limit:

$$\frac{H(tx)}{H(x)} = \frac{H(tx)}{\bar{F}(tx)} \cdot \frac{\bar{F}(tx)}{\bar{F}(x)} \cdot \frac{\bar{F}(x)}{H(x)}.$$

Used in: **MP4.4 (Breiman $\Rightarrow$ RV)**; **MP3.2:** $n\bar{F}(a_n x) = \frac{\bar{F}(a_n x)}{\bar{F}(a_n)} \cdot n\bar{F}(a_n)$; **MP5:** $\frac{n}{k} = \frac{1}{\bar{F}(u_n)}$ turns $n \cdot$ into a tail ratio.

D. Inequalities and bounds

Trick 8: Explicit sandwich + squeeze

Find simple functions below and above, e.g. for $x \geq 1$: $x \leq 1 + x \leq 2x$; $1 \leq 2 + \sin(\cdot) \leq 3$; $0 \leq 1 - 1/t \leq 1$. Remember that **negative powers reverse inequalities**: $x \leq 1 + x \leq 2x \Rightarrow (2x)^{-\alpha} \leq (1 + x)^{-\alpha} \leq x^{-\alpha}$.

Used in: Lomax moments (Potter with $\varepsilon = 0$); **EE8** (sandwiched but not RV); **MP1.3** ($|h| \leq \frac{1}{u+1}$).

Trick 9: Monotonicity of $\bar{F}$ does a lot of work

$\bar{F}$ is decreasing, so $a \leq b \Rightarrow \bar{F}(a) \geq \bar{F}(b)$. Hence $\inf_{[a,b]} \bar{F} = \bar{F}(b)$ and $\sup_{[a,b]} \bar{F} = \bar{F}(a)$, with no continuity needed.

Used in: **Ex 1B**: $\bar{F}(x-y) \geq \bar{F}(x)$ and $\bar{F}(x-y) \geq \bar{F}(x-y_0)$; **Ex 6 (Potter lower bound)**: $z^{\alpha+\varepsilon} \bar{F}(z) \geq \varepsilon^{\alpha+\varepsilon} \bar{F}(b')$ on $[\varepsilon, b']$; **Ex 1C**: bridging non-integer z .

Trick 10: Event inclusion + union bound

If $A \subseteq B_1 \cup B_2 \cup B_3$ then $\mathbb{P}(A) \leq \mathbb{P}(B_1) + \mathbb{P}(B_2) + \mathbb{P}(B_3)$. If $A \supseteq B$ then $\mathbb{P}(A) \geq \mathbb{P}(B)$. **How to find the cover**: ask “if none of the B ’s happen, can A still happen?” and show it cannot.

Used in: **Ex 1A (RV $\Rightarrow$ S)**: $\{Z_1 + Z_2 > x\} \subseteq \{Z_1 > (1-\delta)x\} \cup \{Z_2 > (1-\delta)x\} \cup \{\text{both} > \delta x\}$; lower bound $\{Z_1 + Z_2 > x\} \supseteq \{M_2 > x\}$. **MP2.4**: $\{Z_1 > z\} \subseteq \{Z_1 + Z_2 > z\}$.

Trick 11: liminf/limsup sandwich with a free parameter δ

To prove $\lim = 1$: show $\liminf \geq 1$ and $\limsup \leq g(\delta)$ for every δ , with $g(\delta) \rightarrow 1$ as $\delta \rightarrow 0$.

Used in: **Ex 1A**: $\limsup \leq (1-\delta)^{-\alpha}$, then let $\delta \downarrow 0$; **Rep. thm Part 1**: bound by $\eta \log t$, then let $\eta \rightarrow 0$. **Pattern**: “for every $\eta > 0$, eventually $|\dots| \leq \eta \cdot C$ ” means “ $\rightarrow 0$ ”.

Trick 12: Choose $\varepsilon = \text{half the gap}$

If you need $p < \alpha - \varepsilon$ (or $p > \alpha + \varepsilon$), take $\varepsilon = |\alpha - p|/2$. A strict inequality always leaves room for this.

Used in: **Potter moments** (Ex/PB Use 1); **PB Use 4** ($\varepsilon = \frac{\alpha-1}{2}$ for the mean excess); **PB Use 3** (split the gap in three when a log also appears).

Trick 13: Iterate a one-step bound

If $a_{n+1} \geq q a_n$ then $a_n \geq q^n a_0$. Combine with a growth factor to get geometric blow-up.

Used in: **Ex 1C (L $\Rightarrow$ HT)**: $\bar{F}(z_0+n) \geq (1-\eta)^n \bar{F}(z_0)$, so $e^{t(z_0+n)} \bar{F}(z_0+n) \geq C[(1-\eta)e^t]^n \rightarrow \infty$.

E. Integrals

Trick 14: The power rule (the one integral you must know)

$$\int_1^\infty x^r dx < \infty \iff r < -1 \left(= \frac{1}{-r-1} \right), \quad \int_0^1 x^r dx < \infty \iff r > -1.$$

Useful special case: $\int_1^\infty t^{-\beta-1} dt = \frac{1}{\beta}$.

Used in: Everywhere: moments, $\gamma = \int_1^\infty t^{-1/\gamma-1} dt$, Hill bias, Potter tails. **Always reduce to this.**

Trick 15: Tail formula: expectation = integral of the tail

$\mathbb{E}g(Z) = g(0) + \int_0^\infty g'(x)\bar{F}(x) dx$ for $Z \geq 0$ and increasing, differentiable g .

Why: $g(z) - g(0) = \int_0^\infty g'(x)\mathbf{1}(x < z)dx$; take $\mathbb{E}$ and swap by Tonelli.

Used in: $\mathbb{E}Z^p$ (moments), $\mathbb{E}e^{tZ}$ (Ex 4), $\mathbb{E}\log(Z/u \mid Z > u) = \int_1^\infty \frac{\bar{F}(us)}{\bar{F}(u)} \frac{ds}{s}$ (Hill target), mean excess (PB Use 4).

Trick 16: Split the integral where the behaviour changes

$\int_0^\infty = \int_0^1 + \int_1^\infty$: near 0 use trivial bounds ($\bar{F} \leq 1$), and in the tail use the real tool (Potter, RV).

Used in: Moments (split at 1); **Ex 1B** (split at y_0); **Ex 6** (three regions $(0, 1)$, $[1, b']$, $[b', \infty)$).

Trick 17: Substitutions that simplify

- $u = xs$: turns $\int_x^{tx}$ into $\int_1^t$, a *fixed* range. Used in the representation theorem and von Mises.
- $s = e^v$ ($ds/s = dv$): turns $\int_1^\infty \cdot \frac{ds}{s}$ into $\int_0^\infty \cdot dv$. Used in Ex 3 ($\gamma = \mathbb{E}\log Y$) and the Rep. thm Part 2.
- $v = T(t) = t^{-1/\gamma}$ ($\frac{dt}{t} = -\gamma \frac{dv}{v}$): used for the Hill variance (MP5.3).

Tip: measures of the form dt/t are scale-invariant, so they love multiplicative substitutions.

Trick 18: Uniform smallness lets you integrate an $O(\cdot)$

If $|h(t)| \leq \delta_u$ for all t (with $\delta_u \rightarrow 0$), then $O(h^2) \leq C\delta_u^2$ uniformly, and $\int O(h^2)w(t)dt = O(\delta_u^2)$ for integrable w .

Used in: MP1.3: $|h| \leq \frac{1}{u+1}$ for all $t \geq 1$, so the remainder integrates to $O((u+1)^{-2}) = O((k/n)^{2\gamma})$.

F. Limits and probability

Trick 19: Dominated convergence with a Potter dominating function

To show $\int f_u \rightarrow \int f$ (or $\mathbb{E}f_u(W) \rightarrow \mathbb{E}f(W)$): (1) find the pointwise limit (regular variation); (2) bound $|f_u| \leq g$ with g integrable and *independent of u* (Potter v2); (3) apply DCT.

Used in: Breiman (MP4); mean excess (PB Use 4); **Hill target** $\mathbb{E}[\log(Z/u) \mid Z > u] \rightarrow \gamma$; **Thm 3.13** in the notes.

Trick 20: Mean $\rightarrow$ target and variance $\rightarrow 0 \Rightarrow$ convergence in probability

By Chebyshev, $\mathbb{P}(|X_n - \mathbb{E}X_n| > \eta) \leq \text{Var}X_n/\eta^2$. So if $\mathbb{E}X_n \rightarrow c$ and $\text{Var}X_n \rightarrow 0$, then $X_n \xrightarrow{P} c$.

Used in: Thm 3.9 / MP5.1: $\tilde{T}_n(s) \xrightarrow{P} s^{-\alpha}$. **Sibling:** Markov, $\mathbb{E}e^{tZ} \geq e^{tz}\mathbb{P}(Z > z)$ (Ex 1C, heavy tails).

Trick 21: Sums of indicators are binomial

$\sum_{i=1}^n \mathbf{1}(Z_i > c) \sim \text{Bin}(n, \bar{F}(c))$: the mean is $n\bar{F}(c)$ and the variance is $n\bar{F}(c)(1 - \bar{F}(c))$.
Covariance of two indicator sums: $n[\bar{F}(c \vee d) - \bar{F}(c)\bar{F}(d)]$.

Used in: MP5.1–5.2 ($\tilde{T}_n$ mean, variance, covariance), Thm 3.9–3.10.

Trick 22: Condition on the other variable

For independent Z, W : $\mathbb{P}(Z + W > x) = \mathbb{E}[\bar{F}(x - W)]$ and $\mathbb{P}(ZW > x) = \mathbb{E}[\bar{F}(x/W)]$.

Used in: Ex 1B: $\mathbb{P}(Z_1 + Z_2 > x) = \bar{F}(x) + \int_0^x \bar{F}(x - y)dF(y)$; MP4 (Breiman).

Trick 23: Brownian motion calculations

$\mathbb{E}[W(a)W(b)] = a \wedge b$, and $\mathbb{E}[\int f(v)W(v)dv \cdot \int g(w)W(w)dw] = \iint f(v)g(w)(v \wedge w)dv dw$. For a double integral with $v \wedge w$ or $v \vee w$, use symmetry: $\iint_{[0,1]^2} = 2 \int_0^1 \int_0^w$.

Used in: MP5.3: $\text{Var}(\int_0^1 \frac{W(v)}{v} dv - W(1)) = 2 - 2 + 1 = 1$, so the Hill variance is γ^2 .

Trick 24: Solving rate conditions: exponent algebra

To solve $\sqrt{k}(k/n)^\gamma \rightarrow 0$ for k : collect powers, $k^{1/2+\gamma}n^{-\gamma} \rightarrow 0 \iff k = o(n^{\gamma/(1/2+\gamma)}) = o(n^{2\gamma/(1+2\gamma)})$. **Tip:** write $k = n^\beta$, plug in, and ask for which β the exponent is negative.

Used in: MP1.4 (condition (3.7)); Ex 10 of the notes has the same flavour.

G. Quick self-test (which tricks?)

- Q1.** Show $\frac{1-(1-x^{-2})^3}{3x^{-2}} \rightarrow 1$. *(binomial expansion)*
- Q2.** Show $(1 - \frac{2}{n} + \frac{1}{n^2})^n \rightarrow e^{-2}$. *$((1 - y_n/n)^n$ with $y_n = 2 - 1/n$)*
- Q3.** Show $\mathbb{E}[Z^p \log Z] < \infty$ for $\text{Pareto}(\alpha)$ and $p < \alpha$. *(tail formula, $\log \leq Cx^\varepsilon$, power rule)*
- Q4.** Show $\frac{(1+2x)^{-3}}{x^{-3}} \rightarrow \frac{1}{8}$ and state the index. *(factor out dominant term)*
- Q5.** Show $x^2 \exp(-x^{0.3}) \rightarrow 0$. *(growth hierarchy: take logs)*